PC SYSTEM FUNDAMENTALS, DISK SECTORS AND PARTITION ANALYSIS

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INTRODUCTION

The purpose of this laboratory exercise was to understand the basic components of a computer system and how its storage devices are organised.

The practical involved examining system specifications, running processes, disk information, partitions and file systems. I used Windows and Kali Linux environments to carry out different system inspection activities.

I also practised identifying disk sectors, calculating partition sizes and understanding how a computer starts up. The exercise introduced me to different layers of digital evidence and how they can be collected during a forensic investigation.

All disk inspection activities were performed using read-only commands to avoid modifying the original data. This practical helped me understand the importance of accurate system documentation and evidence preservation in digital forensics.

TOOLS USED

The following operating systems, commands and software were used during the practical:

Operating Systems

- Windows 10 Home Virtual Machine.
- Kali Linux Virtual Machine.
- Physical laptop running Windows.

Software and Applications

- Oracle VirtualBox.
- Windows PowerShell.
- Windows Command Prompt.
- Disk Management.
- DiskPart.
- Linux Terminal.

Commands Used

For Windows commands:

- systeminfo
- msinfo32
- Get-Disk
- Get-Partition
- Get-Volume
- Get-Process
- tasklist
- diskpart
- list disk
- list partition

For Linux commands:

- lsblk
- fdisk -l
- parted -l
- blkid
- df -h
- df -Th
- date +%s

These commands helped me inspect system information, running processes, storage devices, partitions, file systems and available disk space.

SYSTEM SPECIFICATION EVIDENCE

Windows System Information

I used the Windows System Information utility (msinfo32) to examine the computer's specifications.

The computer used for this practical is a Windows 10 Home virtual machine running on Oracle VirtualBox. The operating system version is 10.0.19045 Build 19045 and the system uses an x64-based architecture.

The virtual machine is allocated 2 GB of RAM, the available RAM is 693MB and uses an Intel Core i7-10610U processor operating at approximately 2.3 GHz. The system manufacturer is identified as innotek GmbH, while the baseboard manufacturer is Oracle Corporation. The number of cores is 1 and the logical processors is 1. The BIOS Version is VirtualBox, 12/1/2006.

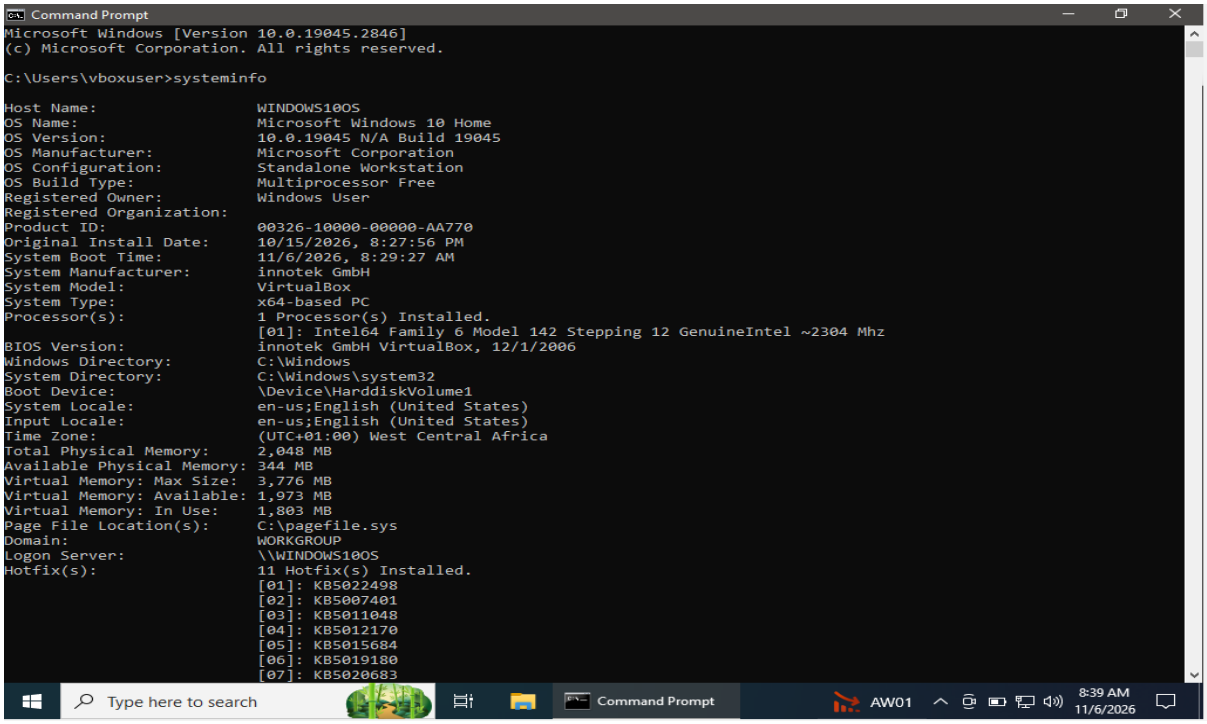
The BIOS mode is Legacy, and the Secure Boot State is reported as Unsupported. These findings provide information about the system's hardware configuration and boot environment.

The System Information utility and the systeminfo command were used to collect the system specifications without modifying the computer.

Observation: The system information confirmed that Windows was running inside VirtualBox as a 64-bit virtual machine. It used Legacy BIOS mode and had 2 GB of allocated memory.

Screenshot Evidence:





Storage Device Information

Storage Information

The storage inspection identified one virtual hard disk named VBOX HARDDISK, with a total capacity of 50 GB. The disk is identified as Disk 0, and its partition style is MBR.

The disk reports a Healthy status and is Online. The partition inspection identified Partition 1, assigned to drive C:, with a starting offset of 1,048,576 bytes and a displayed size of 50 GB.

The volume inspection identified two volumes. Drive C: uses the NTFS file system and has a total capacity of 50 GB, with 21.16 GB of available space. Drive D: is identified as a virtual CD-ROM named VBox_GAs_7.2.12, with a displayed capacity of 51.01 MB.

These commands provided information about the storage devices, partition layout and file systems without modifying the system.

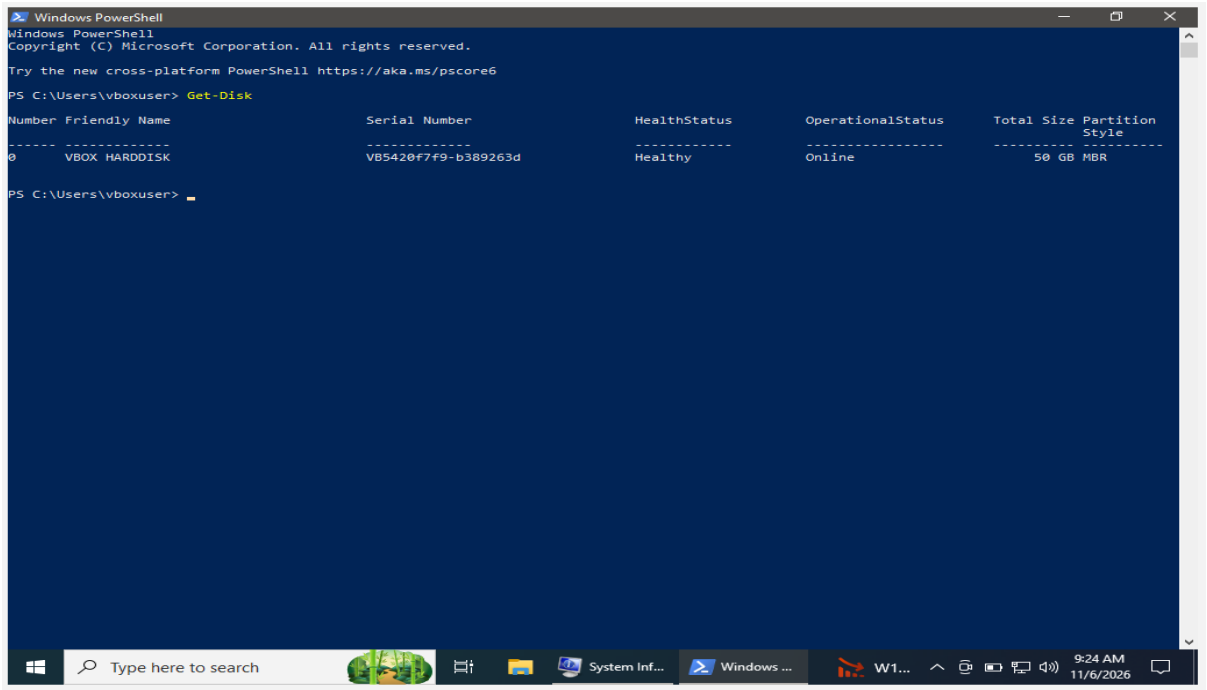
The disk contained one primary partition assigned to drive C:.

The Kali Linux virtual machine contained a virtual disk identified as /dev/sda, with a size of approximately 80.09 GiB.

Observation: The two virtual machines had different disk sizes and partition configurations. Windows used MBR, while Kali Linux also used an MBR partition table.

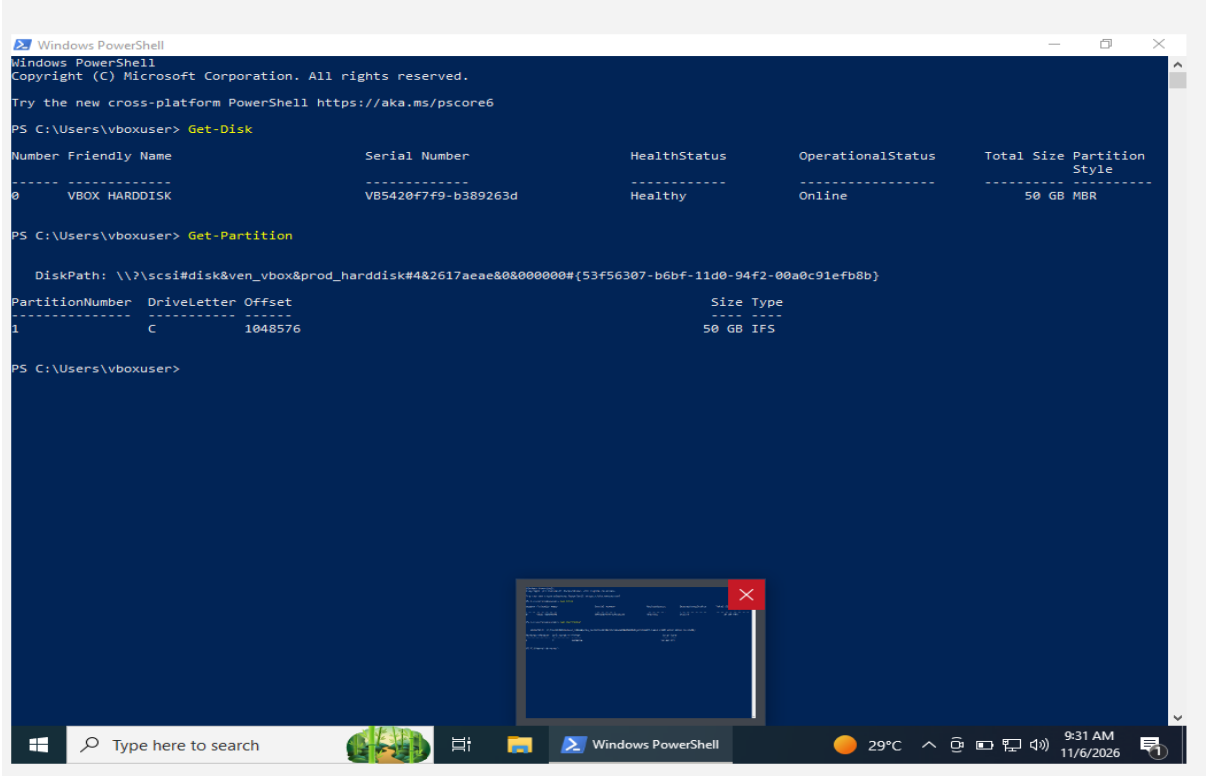
Screenshot evidence:

- Lab1B_A2_Disk_Information.png
- Lab1B_A2_Partition_Information.png
- Lab1B_A2_Volume_Information.png



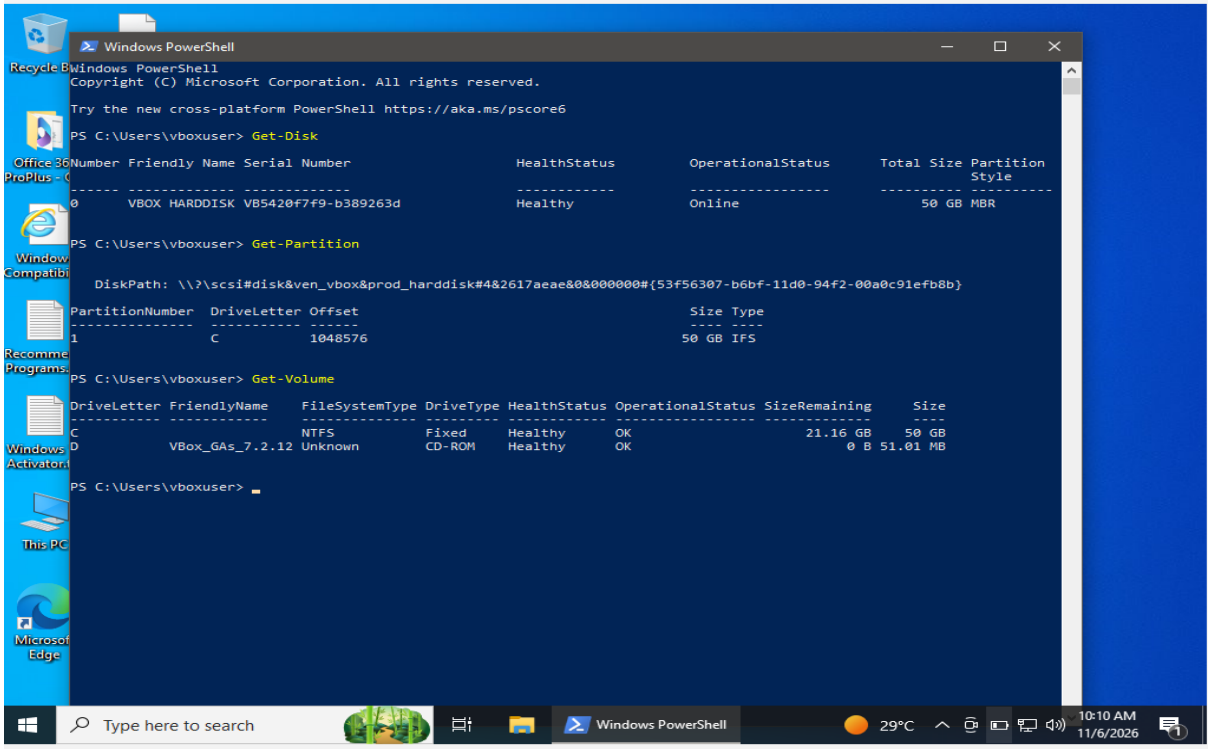
This showed that: Disk Number is 0**,** Serial Number is VB5420f7f9-b389263d, VBOX HARDDISK: Your storage device is a virtual hard disk created by VirtualBox, 50 GB: Your virtual machine has a 50 GB hard disk allocated to it, Healthy: Windows reports that the disk is healthy, Online: The disk is available to the operating system, MBR: Your disk uses Master Boot Record partitioning.

More importantly, My BIOS Mode is Legacy, and my partition style is MBR. These findings are consistent with a traditional Legacy BIOS boot configuration.



This showed that: Disk Number is 0**:** My virtual machine has identified the hard disk as Disk 0**,** Partition Number is 1: Windows has identified one partition on this disk, Drive Letter is C: This is the main Windows partition where your operating system and files are stored, Starting Offset is 1,048,576 bytes**:** This indicates where the partition begins on the disk, The Partition Size is 50 GB (displayed), Partition Type is IFS**:** IFS means Installable File System. It identifies the partition type; it does not tell us the actual file system. We will identify that using Get-Volume.

More importantly, I noticed that my output currently displays only one partition.



This showed that: Volume C is my main Windows storage volume, It uses the NTFS file system, Its total size is 50 GB, It has 21.16 GB of remaining space.

Volume D is a virtual CD-ROM, Its name is VBox_GAs_7.2.12, It is associated with the VirtualBox Guest Additions installation media, Its file system is reported as Unknown.

I observed that my computer has:

- One physical virtual hard disk (Disk 0).
- One partition identified so far (Partition 1).
- Two volumes (C: and D:).

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, shell history	Links user action to system activity
Application layer	Office document metadata, browser cache, chat logs	Application folders and databases	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles	Windows logs, registry, Linux /var/log	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions	File system and unallocated space	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list	Shows live system behaviour

Digital evidence can exist in more than one system layer because computer activities involve different hardware and software components working together.

For example, when a user downloads a file, the user initiates the action, the browser records the download activity, the operating system may record related events and the downloaded file is stored on the disk. Information about the running browser process may also exist in system memory. This means one user activity can leave evidence across several computer system layers.

Therefore, digital forensic investigators examine multiple system layers to obtain a more complete understanding of activities performed on a computer.

Observation: The different evidence layers provide information about how a computer operates. Examining these layers helps an investigator understand the relationship between hardware, the operating system, storage and files.

PART B: PROCESS OBSERVATION

I used the Windows Task List command (tasklist) to examine running processes on the Windows 10 virtual machine.

The tasklist command was executed in Windows Command Prompt to identify processes currently running on the Windows 10 virtual machine.

The command displayed process names, Process Identification Numbers (PIDs), session information and memory usage.

Observation of Running Processes

Three processes were selected for observation:

1. explorer.exe (PID 2908): This process manages the Windows desktop environment, taskbar and File Explorer.

2. MsMpEng.exe (PID 1636): This is a Microsoft Defender Antivirus process responsible for scanning and protecting the system against malware and other security threats.
3. spoolsv.exe (PID 1828): This process manages printing operations and print jobs within Windows.

This observation showed that multiple processes operate simultaneously to provide different system functions. Process information can help forensic investigators examine system activities and identify potentially suspicious processes.

Each process had a unique Process ID (PID), which helps the operating system identify and manage it.

Difference Between Program, Process and Thread

What is a program? A program is a set of instructions stored on a computer that tells the computer what to do. Example: Google Chrome installed on my computer.

WHILE A process is a program that is currently running. Example: When I open Google Chrome, Windows creates a running process. WHILE A thread is a smaller unit of execution within a process. A process can have multiple threads performing different activities.

In summary,

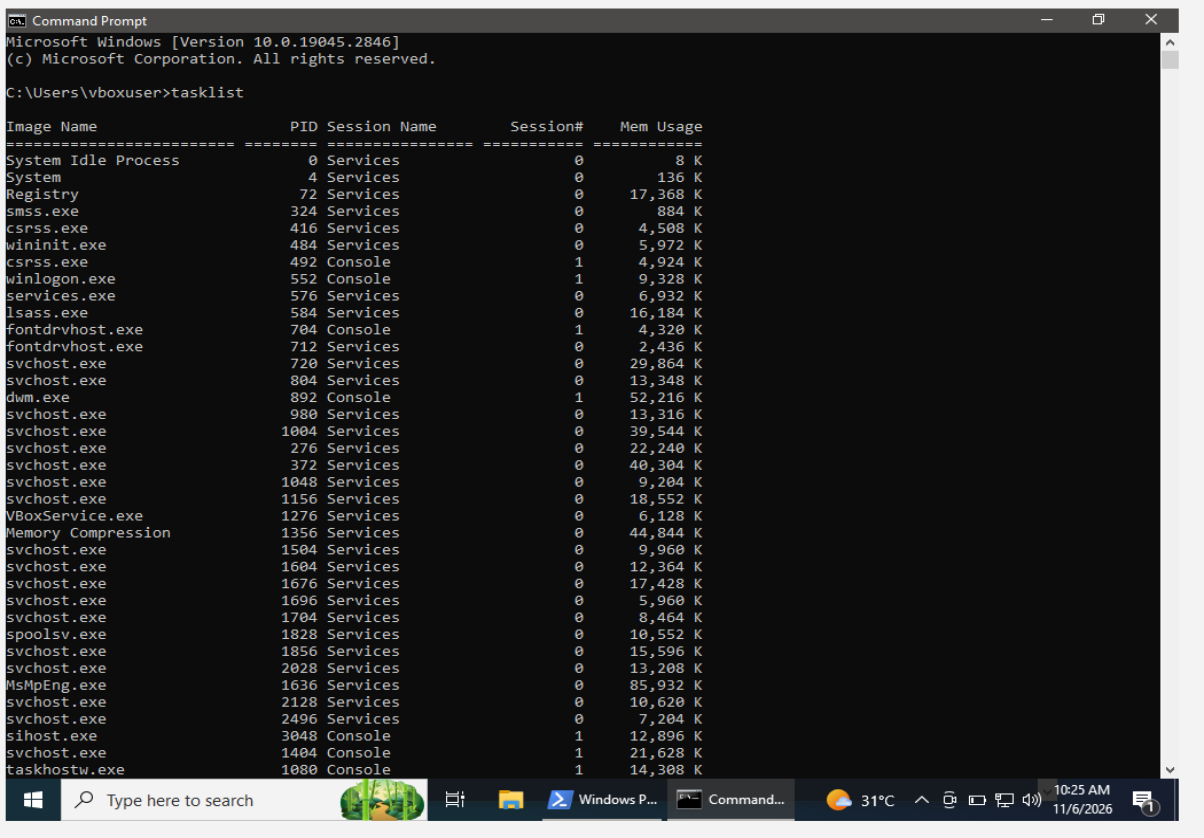
- **Program:** A set of instructions stored on a computer.
- **Process:** A program that is currently running.
- **Thread:** A smaller unit of execution within a process.

Forensic Importance of Process Information

Process information is useful in malware analysis and live forensic investigations because it helps investigators identify programs currently running on a computer. Examining process names, PIDs and memory usage can help identify unusual activities, suspicious software and potentially malicious processes.

Screenshot evidence:

Lab1B_B_Process_Information.png



PART C1.

Calculate the total number of sectors

Formula:

Total sectors = Sectors per track × Heads × Cylinders

Substitute the values:400 × 12 × 17,000

First: 400 × 12 = 4,800

Next: 4,800 × 17,000 = 81,600,000

Answer: 81,600,000 sectors.

Calculate total bytes

Formula: Total bytes = Total sectors × Bytes per sector

Substitute: 81,600,000 × 512

Answer: **41,779,200,000 bytes.**

Convert bytes to Kilobytes (KB)

Formula: KB = Bytes ÷ 1024

Calculation: 41,779,200,000 ÷ 1024

Answer: **40,800,000 KB.**

Convert KB to Megabytes (MB)

Formula: MB = KB ÷ 1024

Calculation: 40,800,000 ÷ 1024

Answer: **39,843.75 MB.**

Convert MB to Gigabytes (GB)

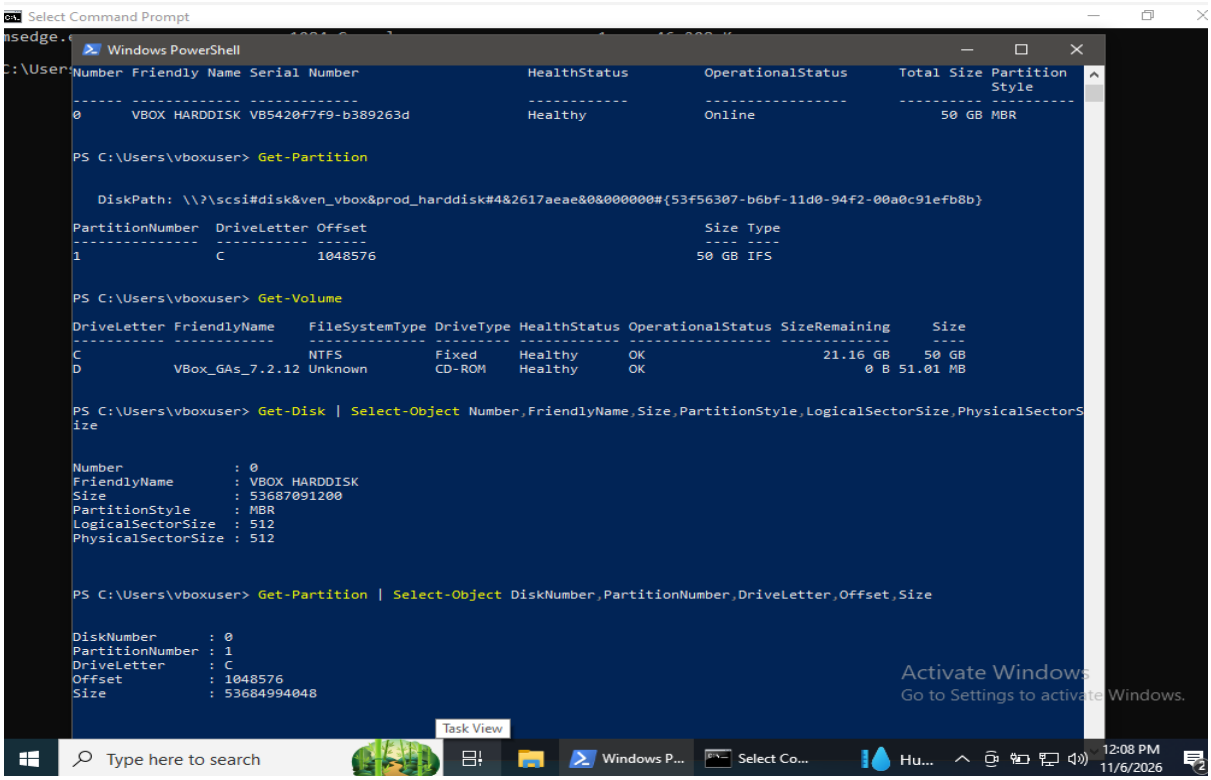
Formula: GB = MB ÷ 1024

Calculation:39,843.75 ÷ 1024

Answer: **38.91 GB.**

FINAL CALCULATION TABLE

Item	Formula / Value	Answer
Total sectors	**400 x 12 x 17000**	**81,600,000 sectors**
Total bytes	**81,600,000 x 512**	**41,779,200,000 bytes**
Kilobytes	**41,779,200,000 / 1024**	**40,800,000 KB**
Megabytes	**40,800,000 / 1024**	**39,843.75 MB**
Gigabytes	**39,843.75 / 1024**	**38.91 GB**



C2: Partition size calculations

My output showed:

Property	Value
Disk Number	0
Partition Number	1
Drive Letter	C:
Starting Offset	1,048,576 bytes
Partition Size	53,684,994,048 bytes
Logical Sector Size	512 bytes

Step 1: Calculate the starting sector using my own disk or partition size

Formula: Starting sector = Offset ÷ Logical sector size

= 1,048,576 ÷ 512

Answer = **2,048**

Step 2: Calculate the number of sectors

Formula: Number of sectors = Partition size ÷ Sector size

= 53,684,994,048 ÷ 512

Answer = **104,853,504 sectors**

Step 3: Calculate the partition size in GB

Formula: Partition size (GB) = Number of sectors × 512 ÷ 1024 ÷ 1024 ÷ 1024

= 104,853,504 × 512 ÷ 1024 ÷ 1024 ÷ 1024

Answer= **49.998 GB (approximately 50 GB)**

Step 4: Calculate the ending sector

Formula: Ending sector = Starting sector + Number of sectors − 1

= 2,048 + 104,853,504 − 1

Answer = **104,855,551**

Part C2 results:

Calculation	Result
Starting sector	2,048
Number of sectors	104,853,504
Ending sector	104,855,551
Partition size	Approximately 50 GB

In summary: The Windows 10 virtual machine contains a 50 GB virtual hard disk identified as Disk 0. The disk uses the MBR partition style, with a logical and physical sector size of 512 bytes. The partition inspection identified one partition, Partition 1, assigned to drive C. Its starting offset is 1,048,576 bytes, while its exact size is 53,684,994,048 bytes.

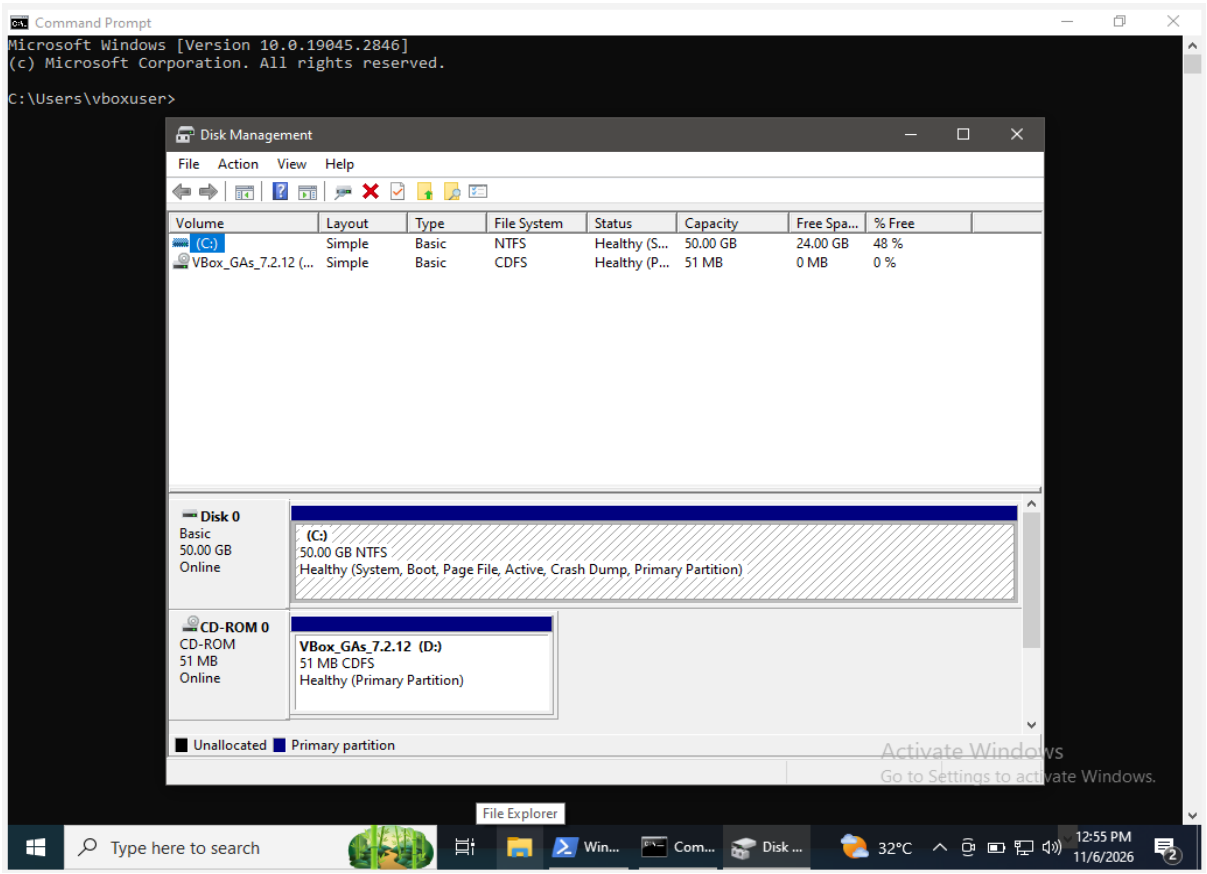
Using a sector size of 512 bytes, the starting sector was calculated as 2,048. The total number of sectors occupied by the partition is 104,853,504, giving an ending sector of 104,855,551. The calculated partition size is approximately 50 GB.

These calculations were performed using the partition information obtained through PowerShell without modifying the disk.

PART D

I opened disk management by pressing Windows + R , then typing diskmgmt.msc and then pressing Enter. I got the output below.

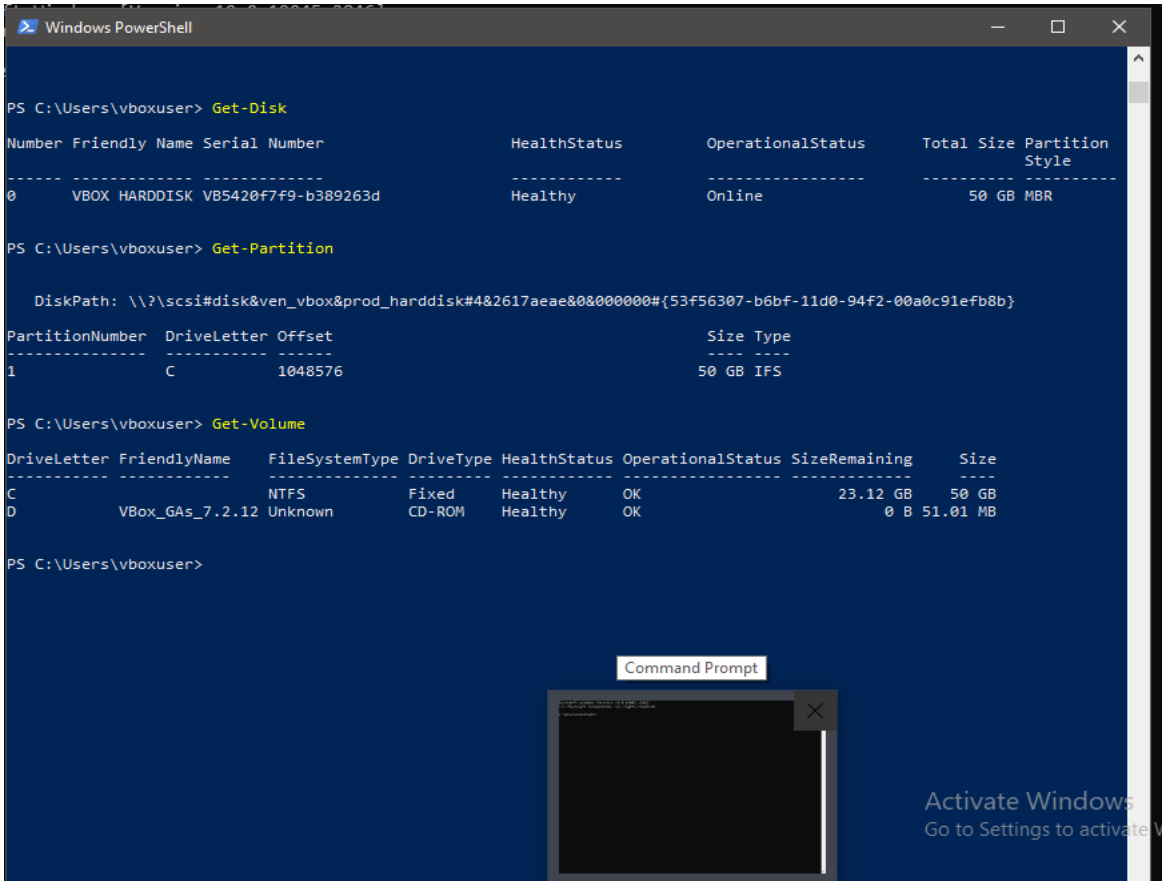
Screenshot: Lab1B_D1_Disk_Management.png



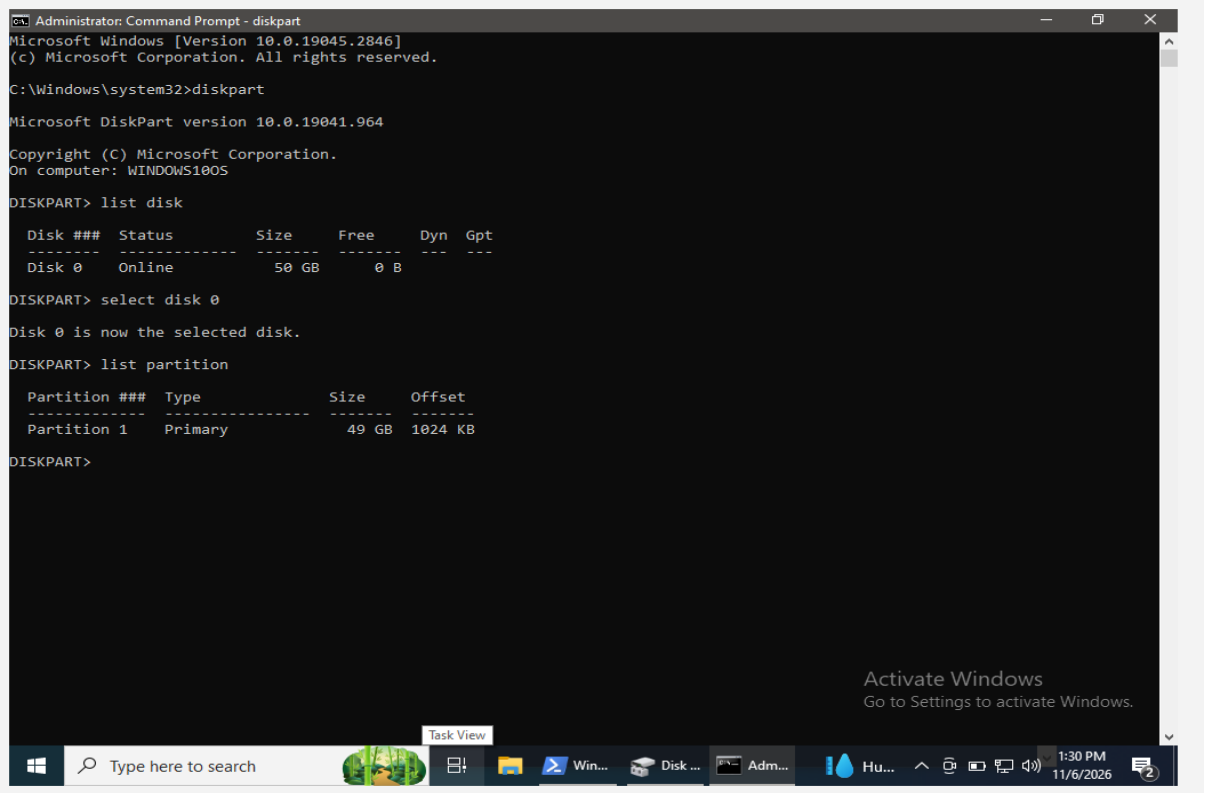
This showed that: Disk 0 is basic, 50 GB, Online, the partition is one primary partition, the Drive C is 50 GB, NTFS, the free space is 24.00 GB, the Drive D is VirtualBox Guest Additions CD-ROM, CD-ROM file system is CDFS, the unallocated space is None visible.

Note that: Drive D: is a CD-ROM, not a second partition on Disk 0. My virtual hard disk has only one partition.

I opened powershell as a user and ran the commands below.



I also opened command prompt as administrator and run DiskPart only for listing disks and partitions using the commands below.

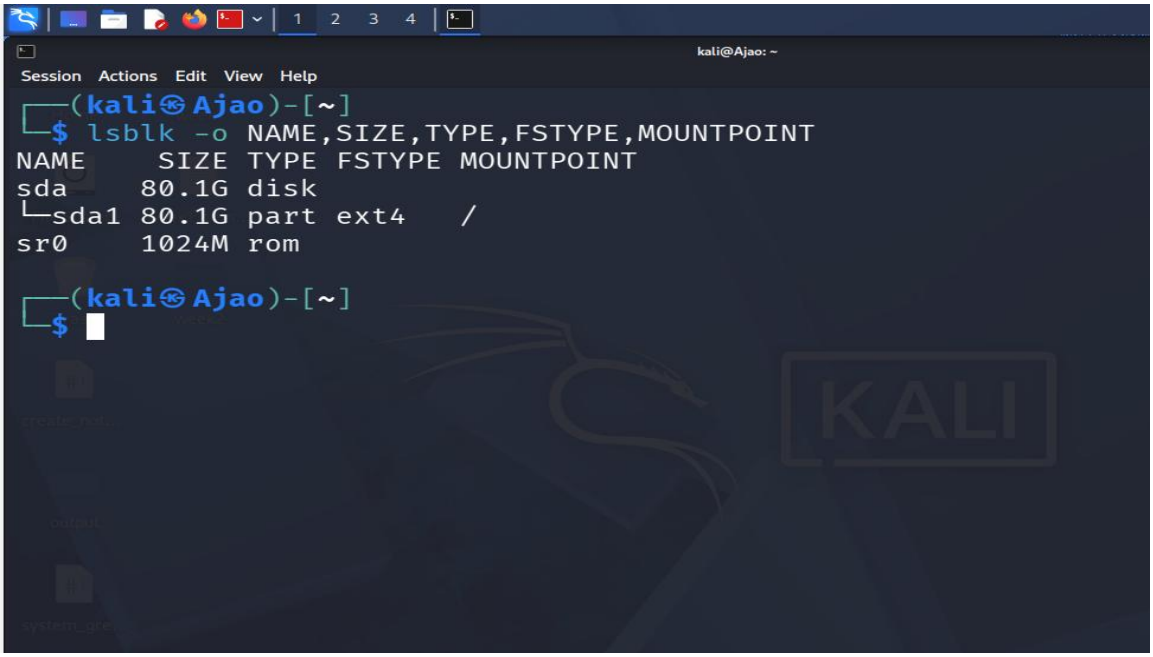




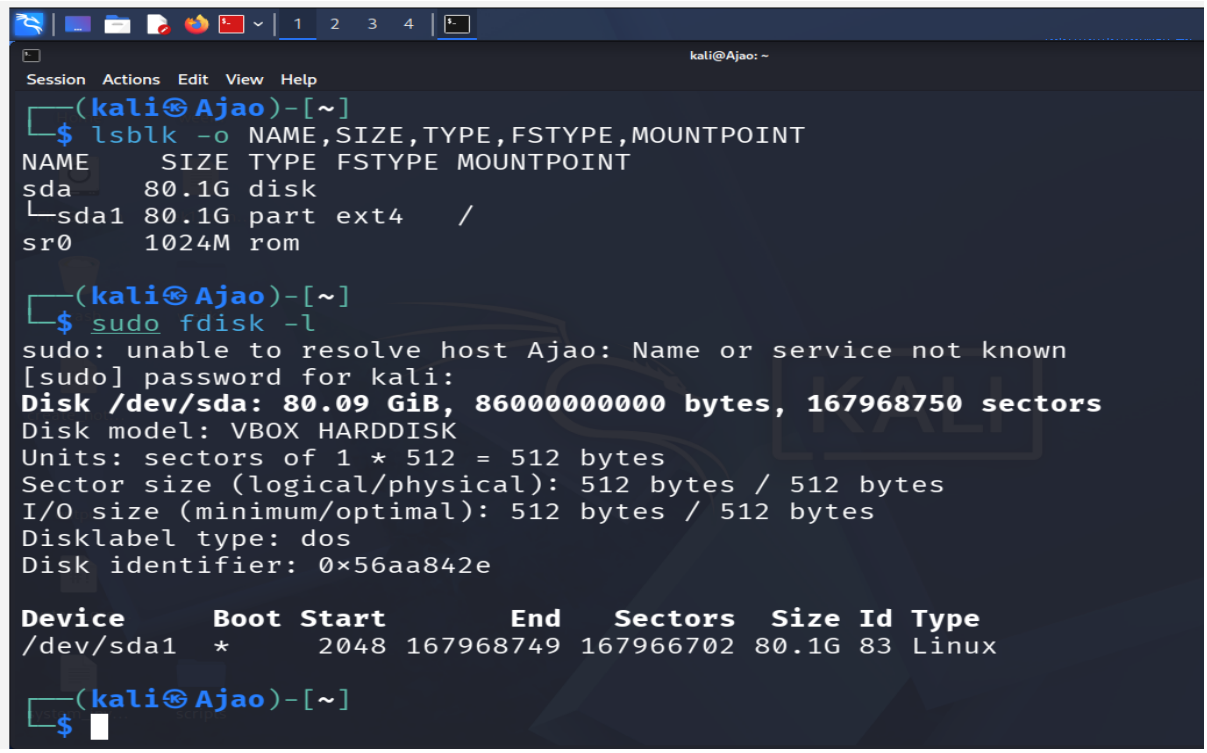
DiskPart was used to inspect the partition table of Disk 0. The disk was reported as Online with a capacity of 50 GB. The partition table identified one Primary partition, Partition 1, with a displayed size of 49 GB and a starting offset of 1024 KB. The disk uses the MBR partition style. No second partition was listed during the inspection. All commands used were for viewing information only, and no changes were made to the disk.

D2: Linux/Kali method

I opened the terminal in my Kali and I ran the commands below and I captured the screenshots.



Name	Size	Type	File System	Mount Point
sda	80.1 GB	Disk	—	—
sda1	80.1 GB	Partition	ext4	/
sr0	1024 MB	ROM/CD-ROM	—	—



This showed that:

/dev/sda is your Kali virtual hard disk.

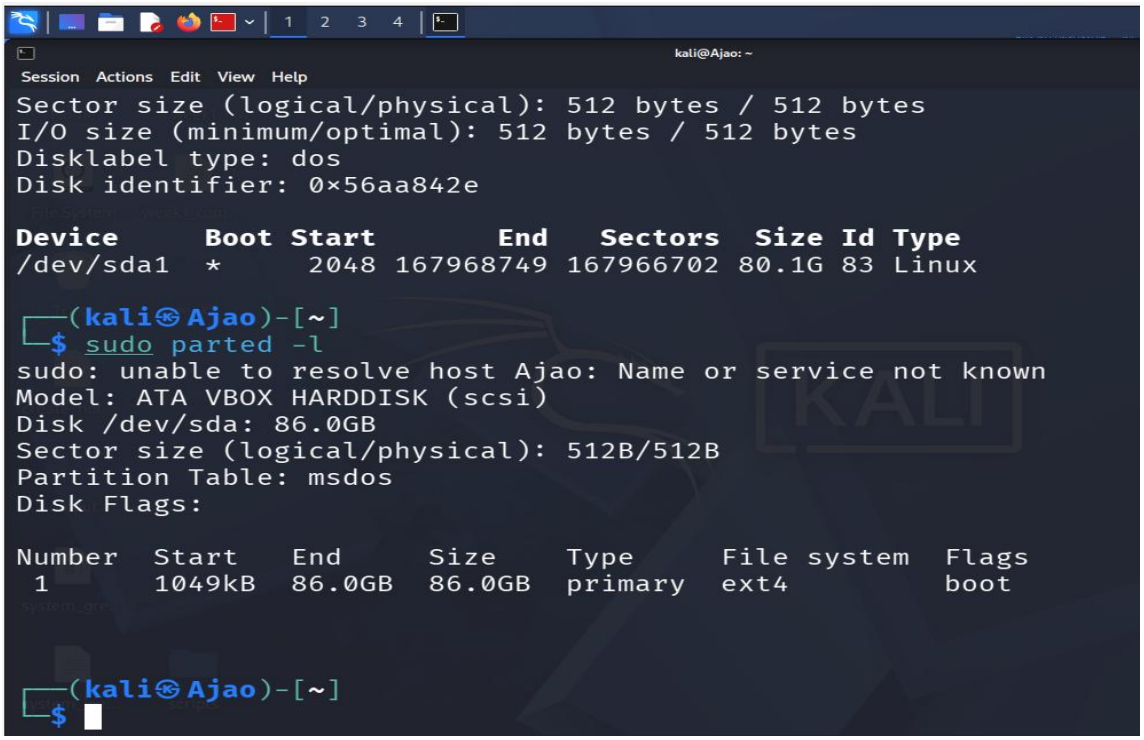
/dev/sda1 is the primary partition on the disk.

ext4 is the Linux file system.

DOS means the disk uses the MBR partition table format.

The * indicates that the partition is marked bootable.

Screenshot: Lab1B_D2_Kali_Fdisk_Partition_Table.png



This showed that:

Kali virtual disk uses the **MBR (msdos) partition table**.

It contains one primary partition.

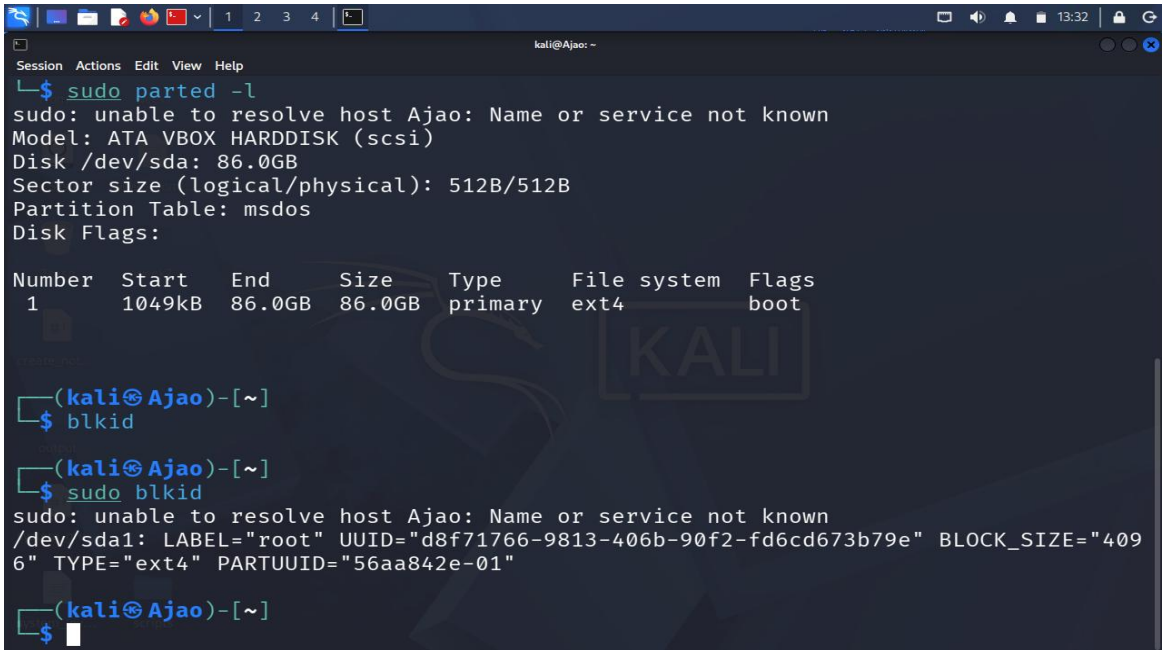
The partition uses the ext4 file system.

The partition is marked as bootable.

The results agree with your earlier fdisk -l inspection.

The slight difference between 80.09 GiB and 86.0 GB is due to the different units used to display disk capacity.

Screenshot filename: Lab1B_D2_Kali_Parted_Disk_Information.png



I used the command sudo parted -l and This showed that:

/dev/sda1 identifies Kali partition.

root is its label.

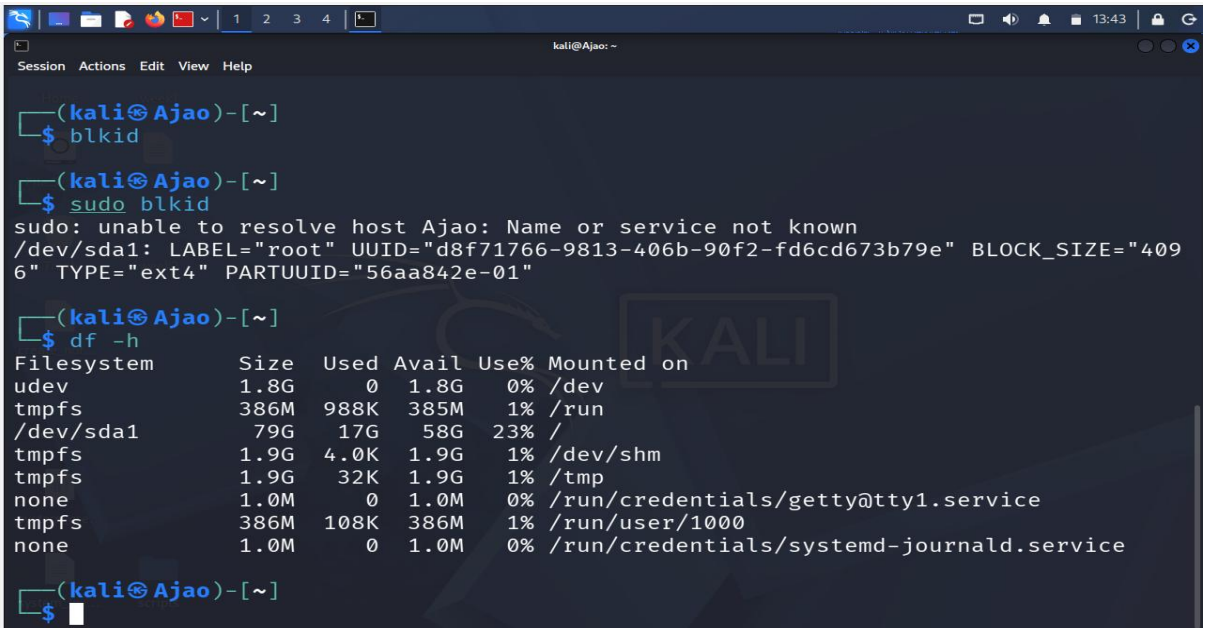
ext4 is the file system used.

UUID is a unique identifier assigned to the file system.

PARTUUID identifies the partition itself.

The block size is 4096 bytes

Screenshot filename: Lab1B_D2_Kali_Blkid_Filesystem_Information.png



I used the command df -h and I got the expected output. The result:

- 79 GB is the total available partition size as displayed by df.
- 17 GB is currently occupied.
- 58 GB is available.
- 23% of the partition is being used.
- / is the root directory of Kali Linux.

The other entries, such as udev and tmpfs, are temporary or memory-backed file systems. They are not additional physical disk partitions.

Screenshot: Lab1B_D2_Kali_Df_Disk_Space.png

In Summary: The Kali Linux virtual machine was inspected using five read-only commands: lsblk, fdisk -l, parted -l, blkid and df -h.

The inspection identified a virtual hard disk named /dev/sda with a capacity of approximately 80.09 GiB. The disk uses the DOS/MBR partition table and contains one primary partition, /dev/sda1. The partition starts at sector 2048 and contains 167,966,702 sectors.

The partition uses the ext4 file system and is mounted at the root directory (/). The blkid command identified its label as root and displayed its UUID and PARTUUID.

The df -h command showed that the partition has approximately 79 GB of usable space, with 17 GB used and 58 GB available. Its reported usage was 23%.

All inspections were performed using read-only commands. No partitions were created, deleted, formatted or modified.

Part E - MBR, GPT and Boot Process Explanation

Stage	What Happens	Forensic Relevance
Power-on and POST	The computer receives power and the system performs the Power-On Self-Test (POST) to initialize and check essential hardware components. Hardware is initialized and checked.	Hardware failure or tampering may affect access to digital evidence.
BIOS/UEFI firmware	BIOS or UEFI initializes hardware and applies firmware settings including	Boot mode and firmware settings help explain the

Stage	What Happens	Forensic Relevance
	the boot device order. CMOS traditionally stores information about firmware settings.	partition style and boot configuration.
MBR/GPT or EFI boot	The system locates the boot information required to start the operating system. Legacy BIOS systems commonly use MBR boot records, while UEFI systems use EFI boot entries and an EFI System Partition.	Boot records may contain evidence of system's boot configuration.
Boot loader/manager	The boot loader or boot manager prepares the operating system for startup and may provide options for selecting an operating system.	Boot configuration may reveal multiple operating systems or unusual boot arrangements.
Kernel and OS control	The operating system kernel loads into memory, followed by essential drivers and services. The operating system then takes control of the computer.	System logs, running processes and user user activities begin to provide evidence for forensic investigation.

Boot process sequence:

Power-on → POST → BIOS/UEFI → MBR/GPT or EFI boot records → Boot loader/manager → Kernel loading → Operating system control.

BOOT PROCESS EXPLANATION

The boot process is the sequence of activities that takes place when a computer is switched on and prepares the operating system for use.

The Windows 10 virtual machine used Legacy BIOS mode and an MBR partition table.

Numbered Boot Sequence

1. **Power On:** The computer is switched on and receives power.

2. **POST:** The BIOS performs the Power-On Self-Test to check essential hardware components.
3. **BIOS Initialisation:** The BIOS identifies available hardware and selects a boot device.
4. **MBR Execution:** The BIOS loads and executes the Master Boot Record from the selected boot disk.
5. **Boot Loader:** The boot loader continues the operating system startup process.
6. **Kernel Loading:** The operating system kernel is loaded into memory.
7. **System Initialisation:** Essential drivers and system services are started.
8. **User Login:** The operating system displays the login screen and becomes ready for use.

Boot mode identified on my system: The Windows 10 virtual machine used for this practical operates in Legacy BIOS mode. This was confirmed using the System Information utility (msinfo32), which displayed BIOS Mode as Legacy. The disk inspection using PowerShell also identified the disk partition style as MBR. These findings are in line with a Legacy BIOS and MBR boot configuration.

Compare MBR and GPT

MBR: Master Boot Record. It is an older partitioning system commonly associated with Legacy BIOS computers.

GPT: GUID Partition Table. It is a modern partitioning system commonly used with UEFI computers.

Feature	MBR	GPT
Age / design	Older partitioning system from the 1980s.	Modern partitioning system used with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures can help investigators recover and validate partition information.

Conclusion:

MBR and GPT are two different partitioning systems used to organize storage devices. MBR is an older system with limitations on disk size and the number of primary partitions. GPT is a newer system that supports larger disks and provides backup partition information. My

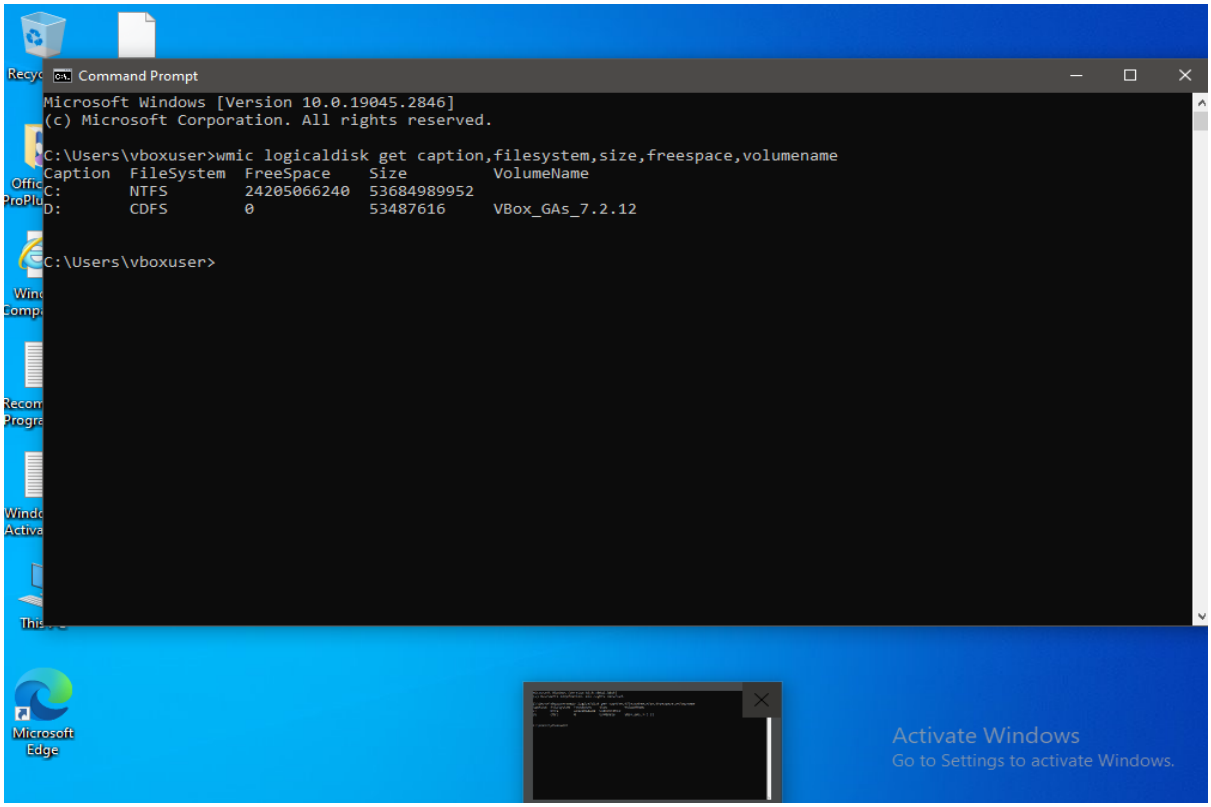
Windows 10 virtual machine uses MBR, as confirmed by the PowerShell and DiskPart inspections.

PART F-File System Identification and Interpretation.

I have already identified three file systems from my earlier practical exercises:

System	Volume/Partition	File System
Windows 10	C:	NTFS
Windows 10	D: (VirtualBox Guest Additions CD-ROM)	CDFS
Kali Linux	`/dev/sda1`	ext4

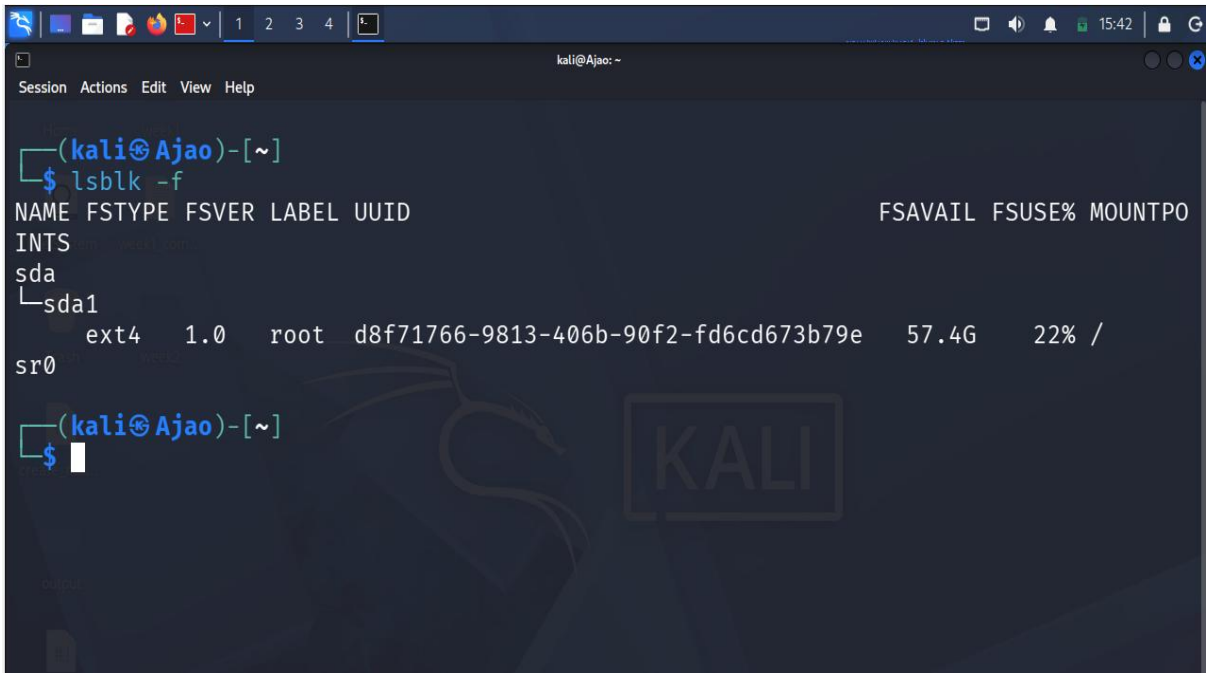
This gives me more than the two file systems required.



For Windows results: My output showed that:

- **C: – NTFS:** This is the main Windows partition. NTFS is the file system used to organise and manage files on the Windows operating system.
- **D: – CDFS:** This is the VirtualBox Guest Additions CD-ROM. CDFS is a file system used for CD-ROM media. It is not a second hard-disk partition.

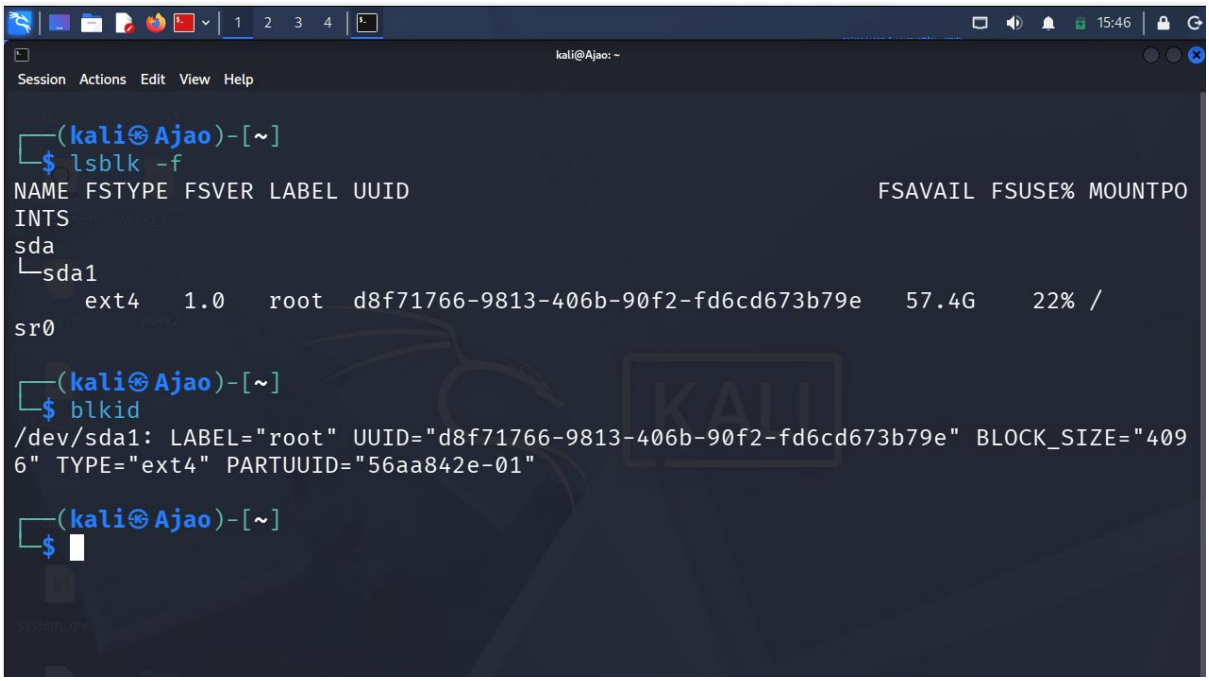
Screenshot filename: Lab1B_F_Windows_File_System_Information.png



For Kali Linux results: The command I used `lsblk -f` output showed:

- `/dev/sda` is virtual hard disk.
- `/dev/sda1` is the partition containing Kali Linux operating system.
- `ext4` is the Linux file system.
- `root` is the partition label.
- `/` means the partition is mounted as the root directory.
- `57.4 GB` is available space.
- `22%` of the partition is currently used.

Screenshot filename: `Lab1B_F_Kali_File_System_Information.png`

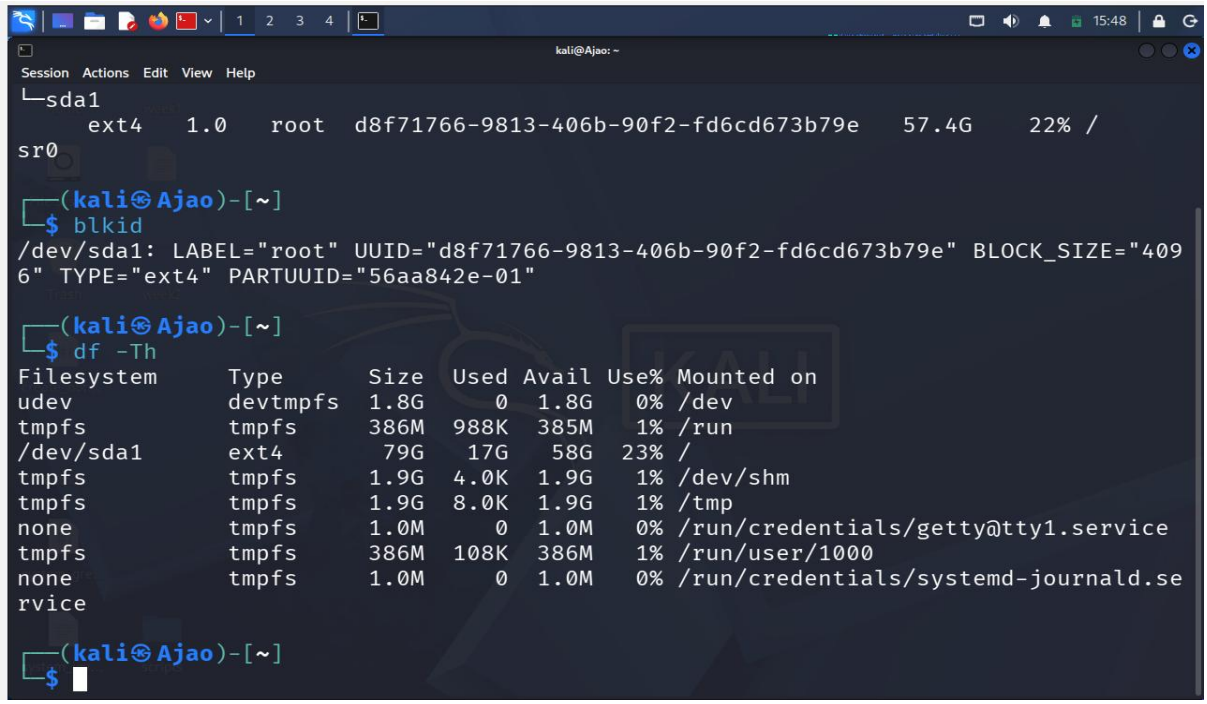


My output confirmed that:

Property	Result
Device	/dev/sda1
File System	ext4
File System Version	1.0 (from lsblk -f)
Label	root
Block Size	4096 bytes
UUID	d8f71766-9813-406b-90f2-fd6cd673b79e
PARTUUID	56aa842e-01

This confirms that my Kali Linux partition uses the **ext4 file system**.

Screenshot filename: Lab1B_F_Kali_Blkid_Filesystem_Information.png



I used the command df -Th and I got the results:

The most important line is:

/dev/sda1 ext4 79G 17G 58G 23% /

Property	Result
Device	/dev/sda1
File system type	ext4
Total size	79 GB
Used space	17 GB
Available space	58 GB
Usage	23%
Mount point	/

The other entries, such as udev and tmpfs, are temporary or memory-backed file systems, not additional physical disk partitions.

Screenshot: Lab1B_F_Kali_Df_Filesystem_Information.png

In Summary, File system inspection was performed on the Windows 10 and Kali Linux virtual machines using read-only commands.

On Windows, the WMIC command and PowerShell Get-Volume command identified the C: drive as NTFS, with a total size of approximately 50 GB. The D: drive was identified as CDFS and represents the VirtualBox Guest Additions CD-ROM.

On Kali Linux, the commands lsblk -f, blkid and df -Th identified the /dev/sda1 partition as using the ext4 file system. The partition is labelled root and is mounted at /. The df -Th command showed approximately 79 GB of total space, 17 GB used and 58 GB available, with 23% usage.

The inspection therefore identified NTFS, CDFS and ext4 across the two virtual machines.

File System	Common OS / Usage	Forensic Importance
FAT/FAT32	Removable media and older systems	Simple structure; often used on USB drives.
NTFS	Modern Windows systems	Includes metadata such as MFT entries, timestamps, permissions, and journals.
ext family	Linux systems	Uses inode-based metadata and supports forensic timeline analysis.

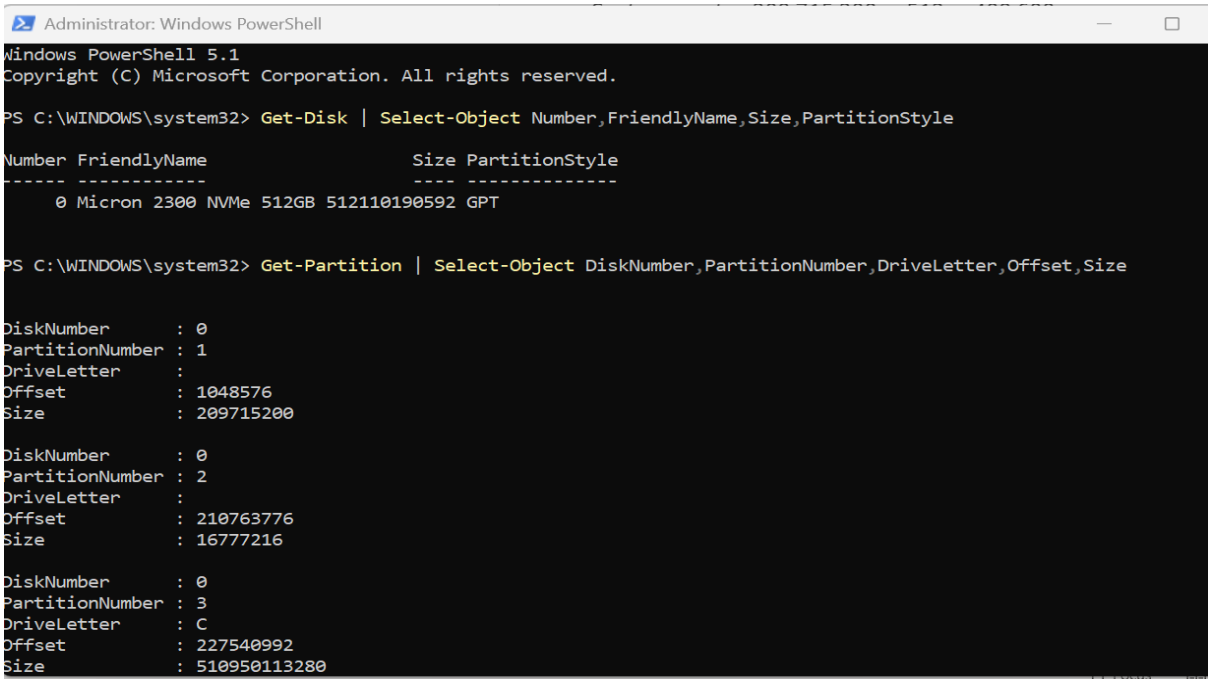
File System	Common OS / Usage	Forensic Importance
XFS/JFS/btrfs	Linux/Unix-like systems	Advanced features may affect recovery and metadata interpretation.
HFS+	Older macOS systems	Important in Mac forensic investigations; slide notes date support limitations.

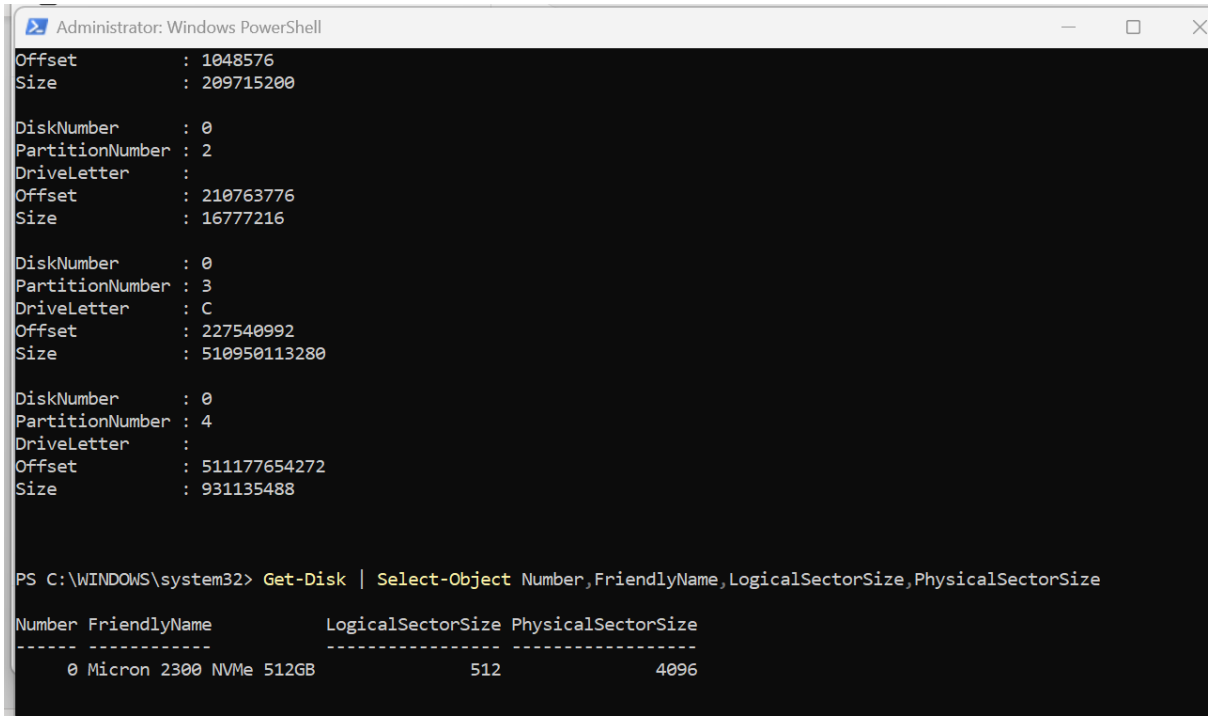
Importance of File Systems in Digital Forensics

File systems are important in digital forensics because they organise how information is stored, accessed and managed on storage devices. They contain metadata such as file names, timestamps, permissions, file locations and allocation information. Investigators use this information to examine user activities, reconstruct timelines, identify deleted files and understand how data was stored or accessed. Different file systems maintain metadata differently, so understanding their structures helps forensic investigators interpret digital evidence accurately.

PART G

I verified the size of partitions 1 and 2 of a PC using a partition table.





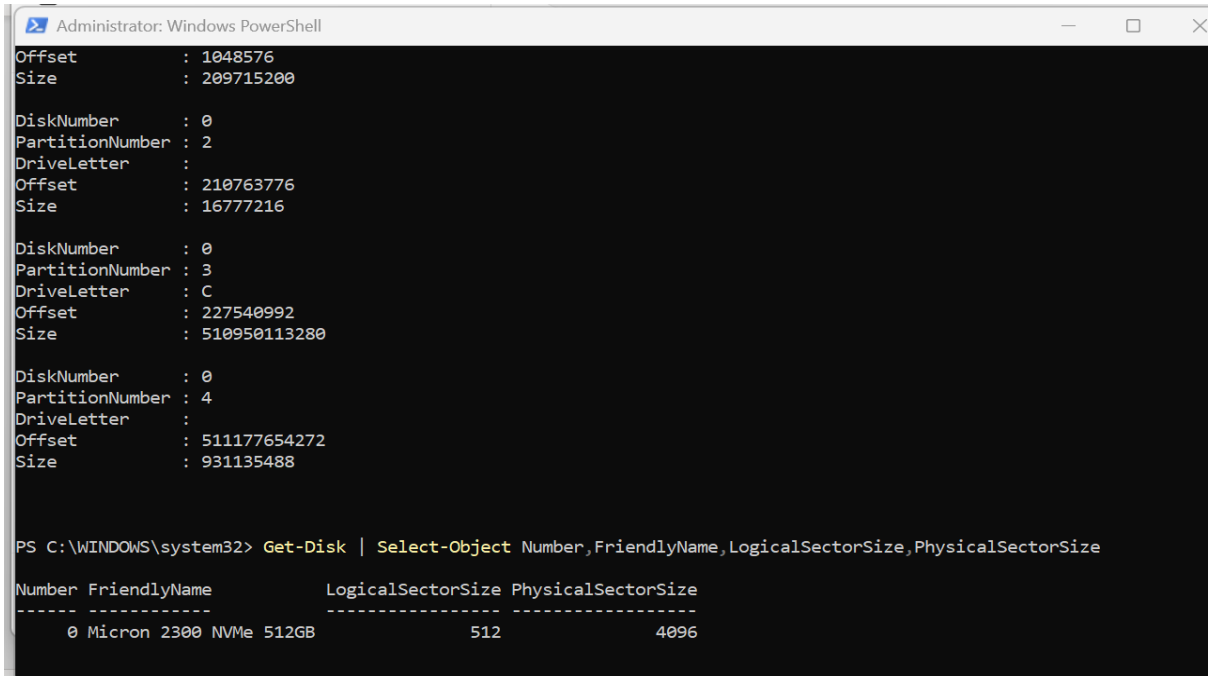
My PC has four partitions on Disk 0, so I used Partition 1 and Partition 2 for this part.

My disk information showed that:

- **Disk:** 0 — Micron 2300 NVMe 512GB
- **Partition style:** GPT
- **Total disk size:** 512,110,190,592 bytes
- **Partition 1:** 200 MB
- **Partition 2:** 16 MB
- **Partition 3:** C: drive
- **Partition 4:** Recovery partition

I used **Partitions 1 and 2 on the same disk**. I didn't create, delete, or modify anything.

I confirmed the sector size, In the same PowerShell window, I ran the command: Get-Disk | Select-Object Number,FriendlyName,LogicalSectorSize,PhysicalSectorSize and I got this output.



My PC uses:

- **Disk:** 0 — Micron 2300 NVMe 512GB
- **Partition table:** GPT
- **Logical sector size:** 512 bytes
- **Physical sector size:** 4096 bytes

For the calculations, I used the **logical sector size of 512 bytes**.

Partition calculations

Information	Partition 1	Partition 2
Partition number	1	2
Start offset (bytes)	1,048,576	210,763,776
Partition size (bytes)	209,715,200	16,777,216
Start sector	2,048	411,648
Sector count	409,600	32,768
End sector	411,647	444,415
Size in GiB	0.1953125	0.015625
Approximate size	200 MiB	16 MiB

How the calculations were done

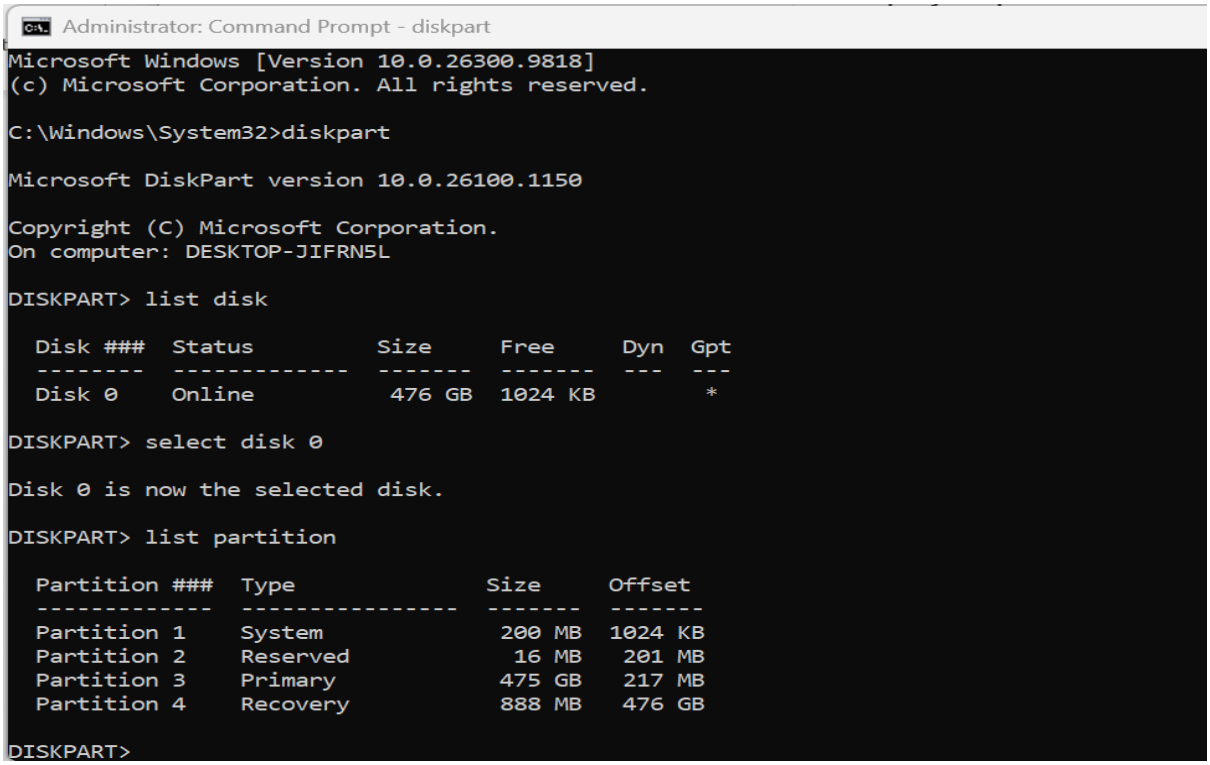
Partition 1

- Start sector = 1,048,576 ÷ 512 = 2,048
- Sector count = 209,715,200 ÷ 512 = 409,600

- End sector = 2,048 + 409,600 − 1 = 411,647

Partition 2

- Start sector = 210,763,776 ÷ 512 = 411,648
- Sector count = 16,777,216 ÷ 512 = 32,768
- End sector = 411,648 + 32,768 − 1 = 444,415



My DiskPart output confirms that my PC has four partitions on Disk 0, and I can clearly identify Partitions 1 and 2.

1. Partition table evidence

My results:

Partition	Type	Size	Offset
Partition 1	System	200 MB	1,024 KB
Partition 2	Reserved	16 MB	201 MB
Partition 3	Primary	475 GB	217 MB
Partition 4	Recovery	888 MB	476 GB

Disk information:

- Disk number: 0
- Status: Online
- Disk size: 476 GB

- Partition style: GPT
- Free space: 1024 KB

2. Calculated results for Partitions 1 and 2

Using my logical sector size of 512 bytes:

Measurement	Partition 1	Partition 2
Start sector	2,048	411,648
End sector	411,647	444,415
Total sectors	409,600	32,768
Size in bytes	209,715,200	16,777,216
Size in GiB	0.1953125	0.015625
Approximate size	200 MiB	16 MiB

These calculations agree with the sizes and offsets shown by DiskPart.

Screenshot: Lab1B_G_Physical_Disk_Partition_Table.png

I made sure the evidence showed the: list disk, select disk 0 and list partition

Partition Size Verification

I inspected the physical laptop's Disk 0 using Windows PowerShell and DiskPart. The disk is a Micron 2300 NVMe 512GB drive with a GPT partition style and a logical sector size of 512 bytes.

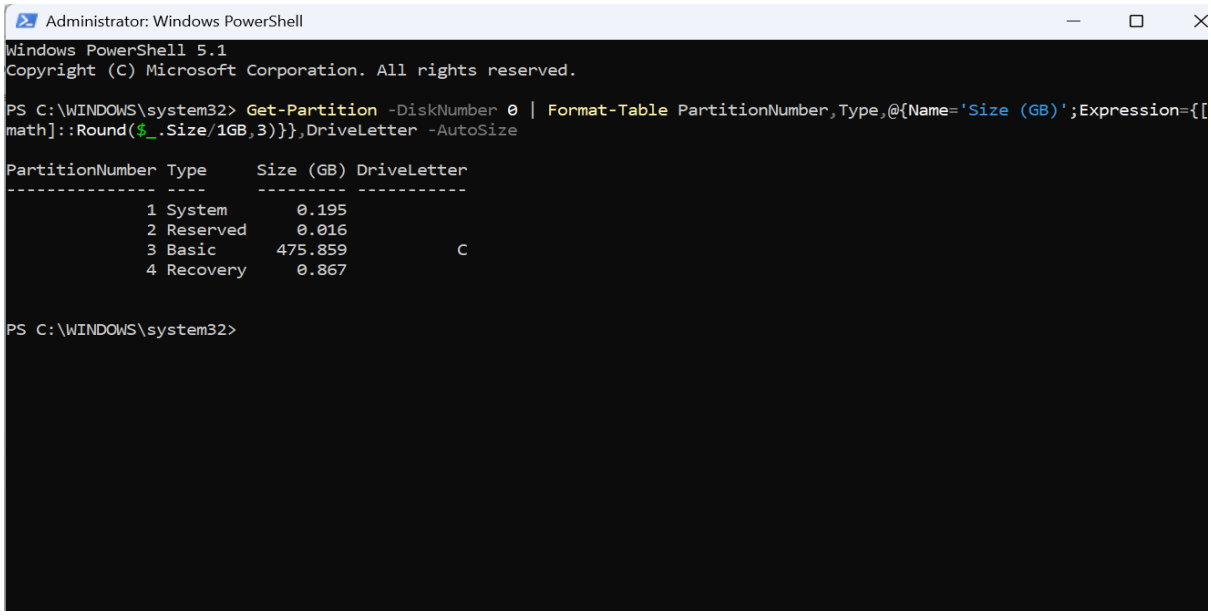
The DiskPart partition table identified four partitions: Partition 1 (System), Partition 2 (Reserved), Partition 3 (Primary), and Partition 4 (Recovery).

Partition 1: The starting offset is 1,048,576 bytes, equivalent to sector 2,048. Its size is 209,715,200 bytes, equivalent to 409,600 sectors. Its ending sector is 411,647. The calculated size is approximately 0.1953 GiB (200 MiB).

Partition 2: The starting offset is 210,763,776 bytes, equivalent to sector 411,648. Its size is 16,777,216 bytes, equivalent to 32,768 sectors. Its ending sector is 444,415. The calculated size is approximately 0.015625 GiB (16 MiB).

The calculated sizes agree with the sizes displayed by DiskPart. Minor differences in displayed values can occur because Windows uses rounded decimal and binary storage units.

The partition-table evidence confirms the positions and sizes of both partitions. All inspection was performed using read-only commands without modifying the disk.



Windows-reported partition sizes

Partition	Type	Windows-reported size
1	System	0.195 GB
2	Reserved	0.016 GB
3	Basic (C:)	475.859 GB
4	Recovery	0.867 GB

I Compared Partitions 1 and 2

Measurement	Partition 1	Partition 2
Partition-table size	200 MiB	16 MiB
Calculated size (GiB)	0.1953125	0.015625
Windows-reported size (GB)	0.195	0.016
Result	Matches	Matches

The small differences are due to rounding and the difference between decimal GB and binary GiB.

I saved the PowerShell screenshot: **Lab1B_G_Windows_Reported_Partition_Sizes.png**

Comparison with Operating-System-Reported Sizes

I used Windows PowerShell to retrieve the partition sizes reported by the operating system. The results showed that Partition 1 (System) has a size of 0.195 GB, while Partition 2 (Reserved) has a size of 0.016 GB.

These values agree with the sizes calculated from the partition-table evidence. Partition 1 has a calculated size of approximately 0.1953 GiB (200 MiB), while Partition 2 has a calculated size of approximately 0.0156 GiB (16 MiB).

The slight differences in displayed values are due to rounding and the use of decimal and binary storage units.

Conclusion for Part G

The inspection confirmed that Disk 0 uses the GPT partition style and contains four partitions. The starting sectors, ending sectors, sector counts and calculated sizes of Partitions 1 and 2 were established using their byte offsets, partition sizes and the logical sector size of 512 bytes.

The calculated partition sizes were consistent with the sizes reported by Windows and DiskPart. This demonstrates how partition-table evidence can be used to verify disk partition information during a forensic examination.

CONCLUSION

This practical helped me understand important computer system components and how digital evidence can be collected from different system layers.

I learned how to inspect system specifications, identify running processes, examine storage devices and determine the file systems used by Windows and Linux.

I also learned how to calculate disk and partition sizes using sector information, offsets and logical sector sizes.

The practical improved my understanding of MBR and GPT partitioning schemes and the sequence of activities involved in the computer boot process.

I also understood that file system information is important when investigating digital evidence because it provides details about how files are stored, accessed and managed.

Most importantly, I learned that digital forensic examinations must be carried out carefully, with proper documentation and without unnecessarily modifying the original evidence.

APPENDIX – ADDITIONAL SCREENSHOTS

The following screenshots provide additional supporting evidence for the practical activities:

Lab1B_A1_System_Information.png

Lab1B_A1_Systeminfo_Output.png

Lab1B_A2_Disk_Information.png

Lab1B_A2_Partition_Information.png

Lab1B_A2_Volume_Information.png

Lab1B_C2_Disk_and_Partition_Information.png

Lab1B_B_Process_Information.png

Lab1B_D1_DiskPart_Partition_Table.png

Lab1B_D1_Disk_Management.png

Lab1B_D2_Kali_Disk_and_Partition_Information.png

Lab1B_D2_Kali_Fdisk_Partition_Table.png

Lab1B_D2_Kali_Parted_Disk_Information.png

Lab1B_D2_Kali_Blkid_Filesystem_Information.png

Lab1B_D2_Kali_Df_Disk_Space.png

Lab1B_G_Physical_Disk_Partition_Table.png

Lab1B_G_Windows_Reported_Partition_Sizes.png